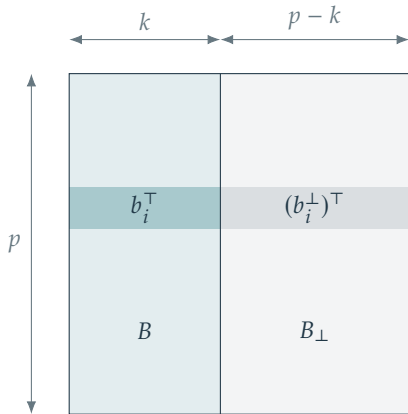




$$\widetilde{B} = [B \ B_\perp]$$

$$\widetilde{B}\widetilde{B}^\top = I_p$$

$$\|b_i\|^2 + \|b_i^\perp\|^2 = 1$$

$$\beta_i = \|b_i\|^2 \leq 1$$